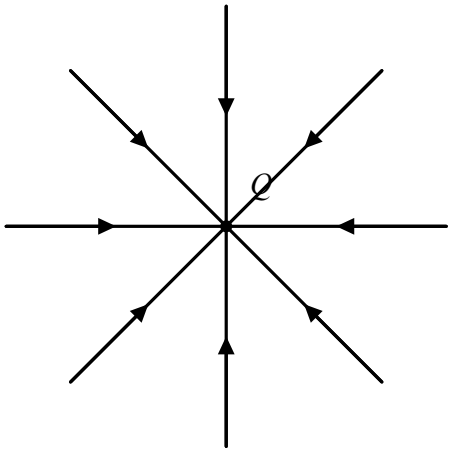
STPM 2014 PHYSICS 960/2 (U2)

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, $g = 9.81$ m s^{-2}, $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m^2 C^{-2}.

Section A [15 marks]

1. The electric field due to a point charge Q is shown in the diagram below. The electric field lines are radial lines which are directed toward the point charge.



Which statement is true about the pattern of electric field?

A The lines are equipotential lines.
B The point charge is positively charged.
C The magnitude of the electric field increases with the increases in distance from the point charge.
D The direction of the lines shows the direction of force experienced by a test positive charge.

Answer: D

Field lines show the direction of the force on a positive test charge (D).

Explanation: The lines point towards Q, so Q is negative; the field weakens with distance; equipotentials are circles.

2. Two point charges attract each other with force F. If one of the charges is tripled and the separation between the charges is reduced to one quarter of its original distance, the new force is

A $\frac{3}{4}F$ B $\frac{9}{16}F$ C $12F$ D $48F$

Answer: D

$$F \propto \frac{Q_1 Q_2}{r^2} : F' = \frac{3}{(1/4)^2} F = 48F$$

Explanation: Quartering the distance multiplies the force by 16.

3. Charges can be induced on a dielectric surface because

A charge carriers in dielectric are electron hole pairs
B electric field cannot pass through the dielectric
C molecules in the dielectric can be polarised
D charges in the dielectric can move freely

Answer: C
The molecules of a dielectric are polarised by a field, giving induced surface charges (C).

Explanation: A dielectric has no free charges; the field does pass through it, reduced.

4. A potential difference of 2.0 V is applied across a wire of resistance 8.2 Ω. What is the number of electrons that flows pass a cross-section in the wire in 30 seconds?

A 1.5×10^{18} B 4.6×10^{19} C 7.7×10^{20} D 3.1×10^{21}

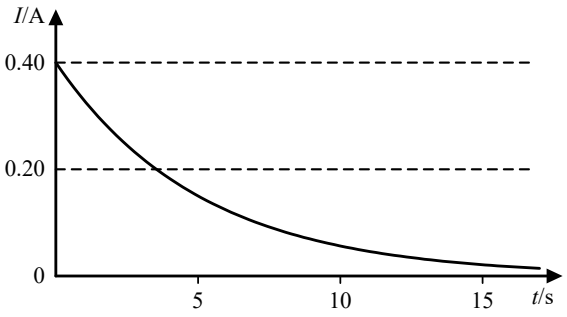
Answer: B

$$I = \frac{2.0}{8.2} = 0.244 \text{ A}; \quad Q = It = (0.244)(30) = 7.32 \text{ C}$$

$$N = \frac{7.32}{1.60 \times 10^{-19}} = 4.6 \times 10^{19}$$

Explanation: Charge = number of electrons × e.

5. A circuit consists of a resistor, a capacitor, a battery and a switch connected in series. When the switch is closed, the variation of current I with time t, is shown in the graph below.



If the e.m.f. of the battery is 10 V, what is the capacitance of the capacitor?

A 75 mF B 100 mF C 200 mF D 1300 mF

Answer: C

$$R = \frac{\varepsilon}{I_0} = \frac{10}{0.40} = 25 \ \Omega$$

The current halves (0.20 A) at about 3.5 s: $\tau = \frac{3.5}{\ln 2} = 5.0$ s; $C = \frac{\tau}{R} = \frac{5.0}{25} = 0.20 \text{ F} = 200 \text{ mF}$

Explanation: The initial current is limited only by R.

6. Bulb P and bulb Q are rated at 60 W, 240 V and 100 W, 240 V respectively. If the filaments of both bulbs are made of the same material and have the same length, the filament of Q has

A lower resistivity
B larger resistance
C lower conductivity
D larger cross-sectional area

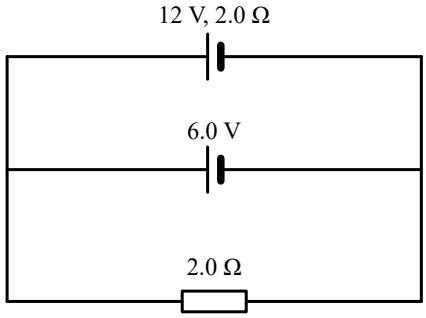
Answer: D

$$R = \frac{V^2}{P}: R_Q = 576\ \Omega < R_P = 960\ \Omega$$

Same material and length: $R = \frac{\rho l}{A}$, so Q has the larger cross-sectional area

Explanation: The resistivity is the same for the same material.

7. A battery of e.m.f. 12 V, internal resistance 2.0 Ω and another battery of e.m.f. 6.0 V without internal resistance are connected in parallel with a resistor of resistance 2.0 Ω as shown in the circuit below.



What is the current flowing through the 6.0 V battery?

A 0 A B 1.0 A C 1.5 A D 3.0 A

Answer: A

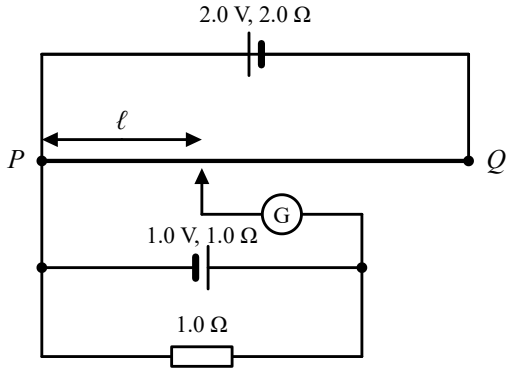
The 6.0 V battery (no internal resistance) fixes the p.d. across the parallel branches at 6.0 V.

Current in the 2.0 Ω resistor = $\frac{6.0}{2.0}$ = 3.0 A; current from the 12 V battery = $\frac{12 - 6.0}{2.0}$ = 3.0 A

So the current through the 6.0 V battery is 3.0 − 3.0 = 0 A.

Explanation: Kirchhoff's first law at a junction.

8. In the potentiometer circuit below, PQ is a slide wire of length 100 cm and resistance 6.0 Ω. The 1.0 V and 2.0 V batteries have internal resistances 1.0 Ω and 2.0 Ω respectively.



What is the balanced length ℓ?

A 25.0 cm B 33.3 cm C 66.7 cm D 75.0 cm

Answer: B

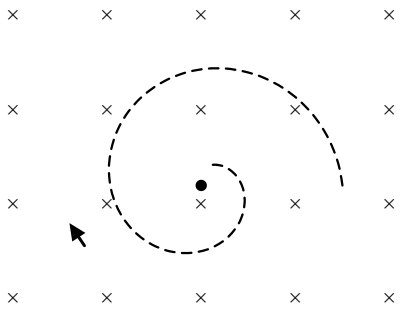
$$\text{Driver current} = \frac{2.0}{6.0 + 2.0} = 0.25 \text{ A; gradient} = \frac{(0.25)(6.0)}{100} = 0.015 \text{ V cm}^{-1}$$

$$\text{P.d. across the 1.0 V cell with the 1.0 Ω load} = 1.0 \times \frac{1.0}{1.0 + 1.0} = 0.50 \text{ V}$$

$$\ell = \frac{0.50}{0.015} = 33.3 \text{ cm}$$

Explanation: The 1.0 V cell supplies current to its 1.0 Ω resistor, so its terminal p.d. is balanced.

9. The track of a charged particle in a magnetic field is shown in the diagram below.



Which statement is true about the particle?

A The particle is a proton which moves with increasing speed.
B The particle is a proton which moves with decreasing speed.
C The particle is an electron which moves with increasing speed.
D The particle is an electron which moves with decreasing speed.

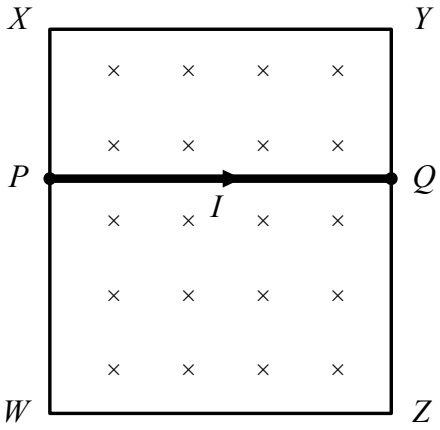
Answer: D

$r = \frac{mv}{qB}$: the radius decreases along the spiral, so the speed decreases

With **B** into the page, the sense in which the particle turns shows a negative charge: an electron (D).

Explanation: A positive charge would turn the other way.

10. A metal rod *PQ* is connected to an e.m.f. so that it remains stationary on a vertical wooden frame *WXYZ* in a uniform magnetic field as shown in the diagram below.



The length of the rod is 40 cm and the mass of the rod is 50 g. If the flux density is 0.20 T, what is the current *I* flowing in the rod?

A 2.5 A B 6.1 A C 15 A D 31 A

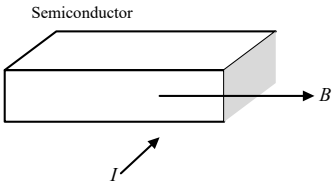
Answer: B

$$BIL = mg \Longrightarrow I = \frac{(0.050)(9.81)}{(0.20)(0.40)} = 6.1 \text{ A}$$

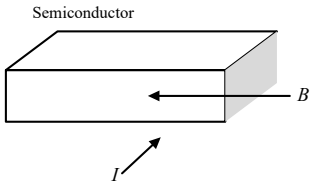
Explanation: The magnetic force supports the weight of the rod.

11. A Hall voltage can be produced when current *I* flows in a thin semiconductor which is placed in a magnetic field *B*. Which diagram will **not** produce Hall voltage?

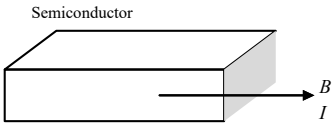
A



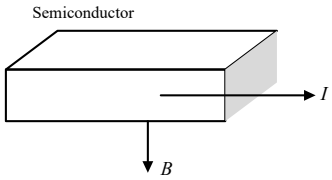
B



C



D

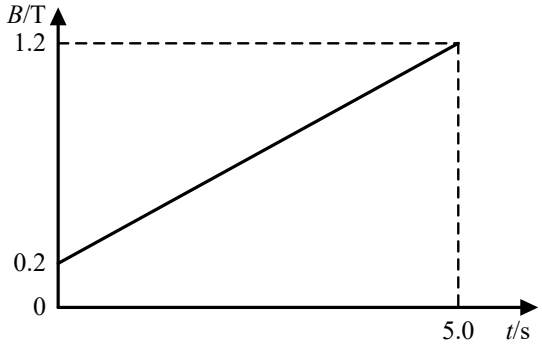


Answer: C

No Hall voltage is produced when **B** is parallel to the current (C), because the force $qv \times \mathbf{B}$ is then zero.

Explanation: **B** must have a component perpendicular to *I*.

12. The graph below shows the variation of a magnetic field *B* with time *t*.



If a circular loop of wire of radius 50 cm is placed in the magnetic field with its plane perpendicular to the field, what is the induced e.m.f. in the loop?

A 0.16 V B 0.20 V C 0.79 V D 0.94 V

Answer: A

$$\frac{dB}{dt} = \frac{1.2 - 0.2}{5.0} = 0.20 \text{ T s}^{-1}; A = \pi(0.50)^2 = 0.785 \text{ m}^2$$

$$\varepsilon = (0.785)(0.20) = 0.16 \text{ V}$$

Explanation: Only the rate of change of *B* matters, not its value.

13. An a.c. power supply is connected in series with a parallel-plate capacitor and an a.c. ammeter. If an insulator is inserted into the space between the plates of the capacitor, the reading of the ammeter will

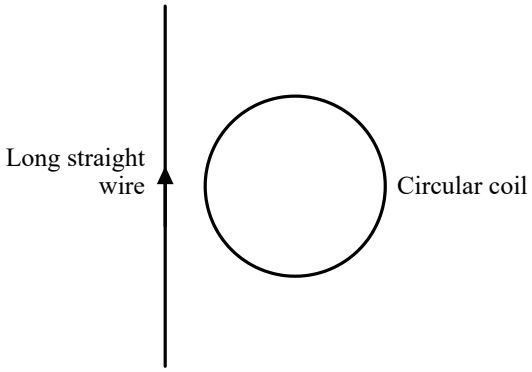
A increase B decrease C become zero D remain constant

Answer: A

The insulator increases *C*, so $X_C = \frac{1}{2\pi fC}$ decreases and the current increases

Explanation: A larger capacitance passes more alternating current.

14. A circular coil and a long straight wire carrying current are fixed near each other is shown in the diagram below.



Which statement is true about the induced current in the coil when the magnitude of the current in the wire is increasing?

A The induced current in the coil flows in the clockwise direction.
B The induced current in the coil flows in the anti-clockwise direction.
C The induced current in the coil initially flows in anti-clockwise direction then in clockwise direction.
D The induced current in the coil initially flows in clockwise direction then in anti-clockwise direction.

Answer: B

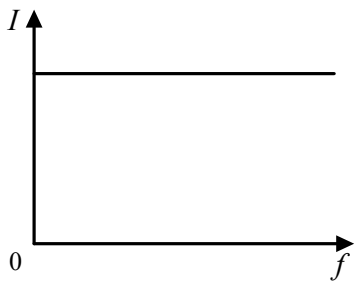
To the right of the upward current, the field of the wire is into the page. As the current increases this flux increases.

The induced current opposes the increase by producing a field out of the page: anticlockwise (B).

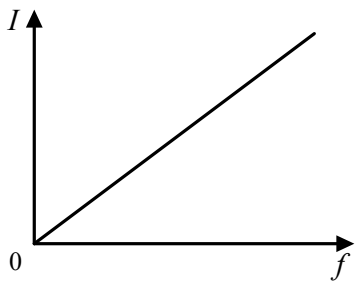
Explanation: Lenz's law.

15. A pure inductor is connected to an a.c. supply. If the frequency of the supply is varied but the amplitude of the voltage remains unchanged, which graph shows the variation of r.m.s. current I with frequency f?

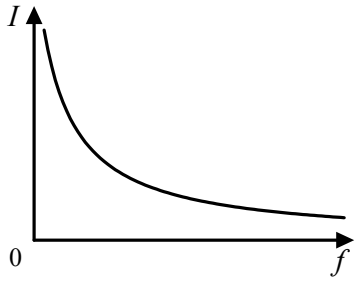
A



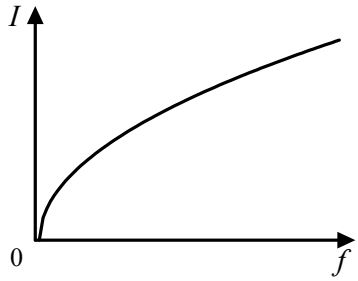
B



C



D



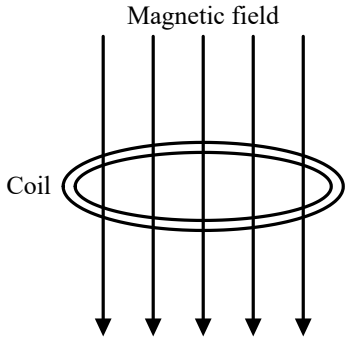
Answer: C

$$I = \frac{V}{2\pi fL} \propto \frac{1}{f}$$: a curve falling towards zero (C)

Explanation: The reactance of an inductor increases with frequency.

Section B [15 marks]

16. A circular coil of 50 turns is placed horizontally as shown in the diagram below.



The radius of the coil is 8.0 cm. A magnetic field of flux density 0.050 T passes perpendicularly through the coil. The magnetic flux density is then increased to 0.100 T in 40 ms.

(a) Indicate the direction of the induced current in the diagram above. (Describe it, e.g. clockwise or anticlockwise when viewed from above.) [1]

Answer

Anticlockwise when viewed from above — 1

1

Explanation: The field points downwards through the coil.

(b) Explain your answer in (a). [2]

Answer

The downward flux through the coil increases	1
By Lenz's law the induced current produces an upward field to oppose the increase, which requires an anticlockwise current viewed from above	1
	2

Explanation: Right-hand grip rule for the induced current.

(c) Calculate the induced e.m.f. [4]

Answer

$A = \pi(0.080)^2 = 2.01 \times 10^{-2}\ \text{m}^2$	1
$\varepsilon = NA\frac{\Delta B}{\Delta t}$	1
$\varepsilon = (50)(2.01 \times 10^{-2})\frac{0.100 - 0.050}{40 \times 10^{-3}}$	1
$\varepsilon = 1.26\ \text{V}$	1
	4
Total for question	**7**

Explanation: The rate of change of B is $1.25\ \text{T s}^{-1}$.

17. (a) State the difference between the power in resistor and pure inductor in an alternating circuit. [2]

Answer

In a resistor V and I are in phase, so energy is dissipated: average power $I_{\text{rms}}^2 R$	1
In a pure inductor V leads I by 90°: energy is stored and returned each cycle, so the average power is zero	1
	2

Explanation: The inductor's instantaneous power alternates in sign.

(b) A coil of wire with inductance 0.125 H and resistance 25.0 Ω is connected to an alternating source of 120 V, 50.0 Hz.

(i) Determine the impedance in the circuit. [4]

Answer

$X_L = 2\pi fL = 2\pi(50.0)(0.125)$ 1

$X_L = 39.3\ \Omega$ 1

$Z = \sqrt{(25.0^2 + 39.3^2)}$ 1

$Z = 46.6\ \Omega$ 1

4

Explanation: The coil's resistance and reactance act in series.

(ii) Calculate the phase angle between the current and the voltage. State whether the current leads or lags the voltage. [2]

Answer

$\tan \varphi = \frac{X_L}{R} = \frac{39.3}{25.0} \Longrightarrow \varphi = 57.5°$ 1

The current lags the voltage 1

2

Total for question **8**

Explanation: In an *RL* circuit the current always lags.

Section C [30 marks]

18. (a) Define *electric field strength* and *electric potential* at a point. [2]

Answer

Electric field strength: the force per unit positive charge at that point, $E = F/q$ 1

Electric potential: the work done per unit positive charge in bringing a positive test charge from infinity to that point 1

2

Explanation: *E* is a vector; *V* is a scalar.

(b) A metal sphere of radius 3.0 cm carries a charge of −12 μC.

(i) Determine the electric field strength at a point 5.0 cm from the centre of the sphere. [2]

Answer

$E = \frac{kQ}{r^2} = \frac{(9.0 \times 10^9)(12 \times 10^{-6})}{(0.050)^2}$ 1

$E = 4.32 \times 10^7$ N C^{-1}, directed towards the sphere 1

2

Explanation: Outside, the sphere acts as a point charge at its centre.

(ii) Determine the electric potential at a point 5.0 cm from the centre of the sphere. [2]

Answer

$V = \frac{kQ}{r} = \frac{(9.0 \times 10^9)(-12 \times 10^{-6})}{0.050}$ 1

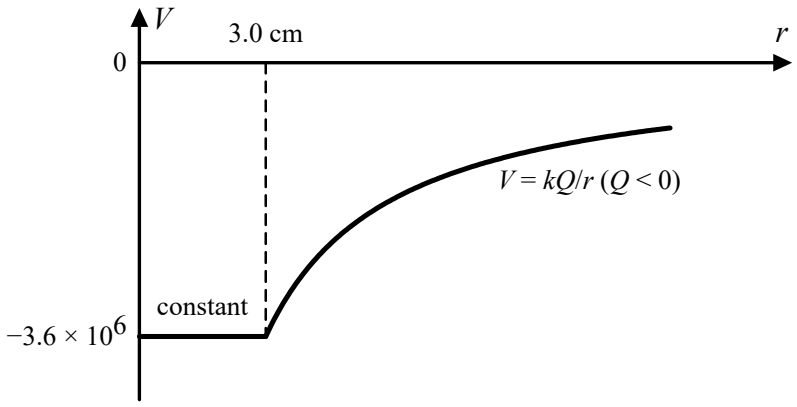
$V = -2.16 \times 10^6$ V 1

2

Explanation: Keep the negative sign for potential.

(iii) Sketch the graph of potential V against distance r from the centre of the sphere. [2]

Answer



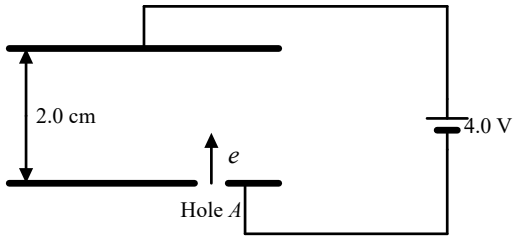
Constant (negative) potential from $r = 0$ to $r = 3.0$ cm 1

For $r > 3.0$ cm, V rises towards zero as $-1/r$ 1

2

Explanation: On the surface $V = -3.6 \times 10^6$ V.

(c) A pair of parallel horizontal metal plates connected to a 4.0 V battery is shown in the diagram below.



The plates are separated at a distance of 2.0 cm. An electron is projected vertically up with a velocity of 9.8×10^5 m s^{-1} through hole *A*.

(i) Determine the distance of the movement of the electron before it comes to rest. [5]

Answer

$E = \frac{V}{d} = \frac{4.0}{0.020} = 200$ V m^{-1} — 1

The electric force on the electron opposes its upward motion: $F = eE$ — 1

Deceleration $a = \frac{eE}{m} = \frac{(1.60 \times 10^{-19})(200)}{9.11 \times 10^{-31}} = 3.51 \times 10^{13}$ m s^{-2} — 1

$v^2 = u^2 - 2as$ with $v = 0$: $s = \frac{u^2}{2a} = \frac{(9.8 \times 10^5)^2}{2(3.51 \times 10^{13})}$ — 1

$s = 1.37 \times 10^{-2}$ m (1.37 cm, less than 2.0 cm) — 1

5

Explanation: Gravity is negligible; energy method: $\frac{1}{2}mu^2 = eEs$ gives the same.

(ii) If the distance between the plates is closer, what is the effect on the electric field and the distance of the electron movement? [2]

Answer

The electric field $E = V/d$ increases — 1

The deceleration increases, so the electron moves a shorter distance before stopping — 1

2

Total for question **15**

Explanation: $s = mu^2d/(2eV) \propto d$.

19. (a) Define *capacitance*. [1]

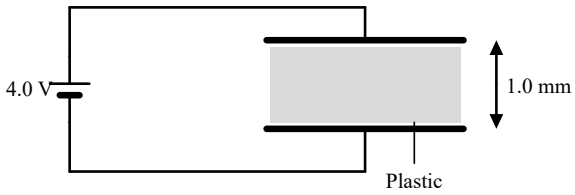
Answer

The ratio of the charge on either plate to the p.d. between them, $C = \frac{Q}{V}$ — 1

1

Explanation: Unit: farad (F).

(b) A parallel-plate capacitor filled with plastic is charged by a 4.0 V battery as shown in the diagram below.



The area of one plate is 200 cm^2. The thickness and dielectric constant of the plastic layer is 1.0 mm and 3.2 respectively.

(i) Calculate the capacitance of the capacitor. [2]

Answer

$C = \frac{\varepsilon_r \varepsilon_0 A}{d} = \frac{(3.2)(8.85 \times 10^{-12})(200 \times 10^{-4})}{1.0 \times 10^{-3}}$ 1

$C = 5.66 \times 10^{-10}$ F 1

2

Explanation: 200 cm^2 = 0.0200 m^2.

(ii) Determine the energy stored in the capacitor. [2]

Answer

$U = \frac{1}{2}CV^2 = \frac{1}{2}(5.66 \times 10^{-10})(4.0)^2$ 1

$U = 4.53 \times 10^{-9}$ J 1

2

Explanation: The average p.d. during charging is half the final p.d.

(iii) Determine the work done by the battery while charging the capacitor. [2]

Answer

$W = QV = CV^2 = (5.66 \times 10^{-10})(4.0)^2$ 1

$W = 9.06 \times 10^{-9}$ J 1

2

Explanation: The battery moves all the charge through its full e.m.f.

(iv) Explain the discrepancy between the answers in (ii) and (iii). [2]

Answer

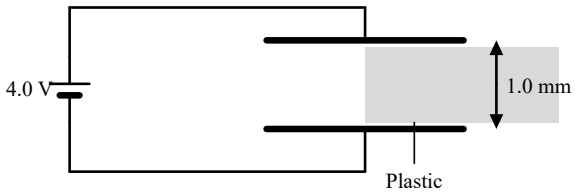
Only half of the work done by the battery is stored in the capacitor — 1

The other half is dissipated as heat in the resistance of the connecting wires (and battery) while the current flows — 1

2

Explanation: This is true whatever the resistance in the circuit.

(c) The battery is removed and the plastic layer is pulled out half from the plates as shown in the diagram below.



(i) Determine the new capacitance of the capacitor. [3]

Answer

Half the area has plastic and half has air: two capacitors in parallel — 1

$C = \frac{\varepsilon_0(A/2)}{d}(3.2 + 1) = \frac{(8.85 \times 10^{-12})(0.0100)}{1.0 \times 10^{-3}}(4.2)$ — 1

$C = 3.72 \times 10^{-10}$ F — 1

3

Explanation: Each half has area 0.0100 m^2.

(ii) Determine the work needed to pull the plastic layer. [3]

Answer

Battery removed: $Q = CV = (5.66 \times 10^{-10})(4.0) = 2.26 \times 10^{-9}$ C stays constant — 1

New energy $U_2 = \frac{Q^2}{2C} = \frac{(2.26 \times 10^{-9})^2}{2(3.72 \times 10^{-10})} = 6.89 \times 10^{-9}$ J — 1

Work = $U_2 - U_1 = 6.89 \times 10^{-9} - 4.53 \times 10^{-9} = 2.36 \times 10^{-9}$ J — 1

3

Total for question **15**

Explanation: The plastic is attracted into the field, so work must be done to pull it out.

20. (a) State Kirchhoff's laws. [2]

Answer

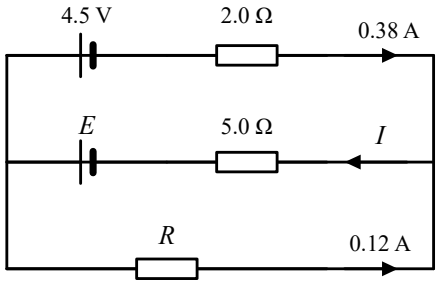
First law: the sum of the currents entering a junction equals the sum leaving it 1

Second law: around any closed loop, the sum of e.m.f.s equals the sum of the *IR* products 1

2

Explanation: Conservation of charge and of energy.

(b) An electric circuit consisting of three resistors and two cells with negligible internal resistance is shown in the diagram below.



Determine the

(i) current *I* which passes through the 5.0 Ω resistor, [1]

Answer

$I = 0.38 + 0.12 = 0.50$ A 1

1

Explanation: Kirchhoff's first law at the right-hand junction.

(ii) e.m.f. *E*, [2]

Answer

Loop through the 4.5 V and *E* branches (the 4.5 V cell opposes *E*): $E = 4.5 + (0.38)(2.0) + (0.50)(5.0)$ 1

$E = 7.76$ V 1

2

Explanation: The 4.5 V cell is being charged by *E*.

(iii) resistance R. [2]

Answer

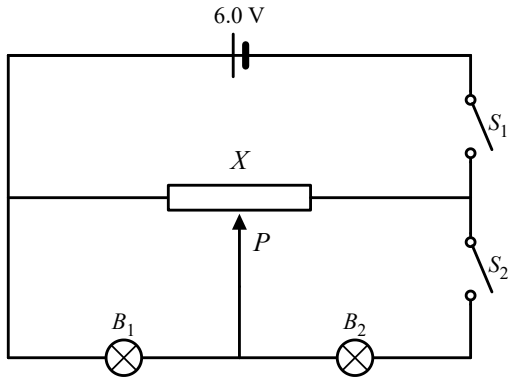
Loop through the E and R branches: $E = (0.50)(5.0) + 0.12R$ 1

$R = \dfrac{7.76 - 2.5}{0.12} = 43.8\ \Omega$ 1

2

Explanation: Check with the outer loop: $4.5 + 0.76 = 0.12R$.

(c) A 6.0 V cell with negligible internal resistance, a potential divider X with slider P, two switches S_1 and S_2 and two identical bulbs B_1 and B_2 are connected as shown in the diagram below.



P is at the centre of the potential divider. The resistance of X is 100 Ω. Bulb B_1 and B_2 each rated 3.0 V, 9.0 W.

(i) Calculate the resistance of the bulb when it functions normally. [2]

Answer

$R = \dfrac{V^2}{P} = \dfrac{3.0^2}{9.0}$ 1

$R = 1.0\ \Omega$ 1

2

Explanation: Rated current = 3.0 A.

(ii) When both switches are closed, what happens to the bulbs? [2]

Answer

The bulbs are in series across 6.0 V and share it equally: 3.0 V each 1

Both light with normal brightness; P is at the same potential as the junction of the bulbs, so no current flows in the slider lead 1

2

Explanation: The two halves of X also divide 6.0 V equally.

(iii) Switch S_2 is then opened, what happens to bulb B_1? Give reasons for your answer. [4]

Answer

With S_2 open, B_2 carries no current; B_1 is in parallel with one half of X (50 Ω) 1

Combined resistance = $\frac{(50)(1.0)}{51}$ = 0.98 Ω, in series with the other 50 Ω half 1

P.d. across $B_1 = 6.0 \times \frac{0.98}{50.98} = 0.12$ V (current about 0.12 A) 1

This is far below 3.0 V (3.0 A), so B_1 does not light 1

4

Total for question **15**

Explanation: The 50 Ω half of the divider limits the current.